Integrated Measuring System - IMS

for Ball and Roller Rail Systems

Linear Motion Technology / DC-LT

Matthias Wolf, DC-LT/SCO3

Integrated Measuring System - IMS
for ball rail and roller rail systems

IMS-A: Integrated Measuring System – Absolute

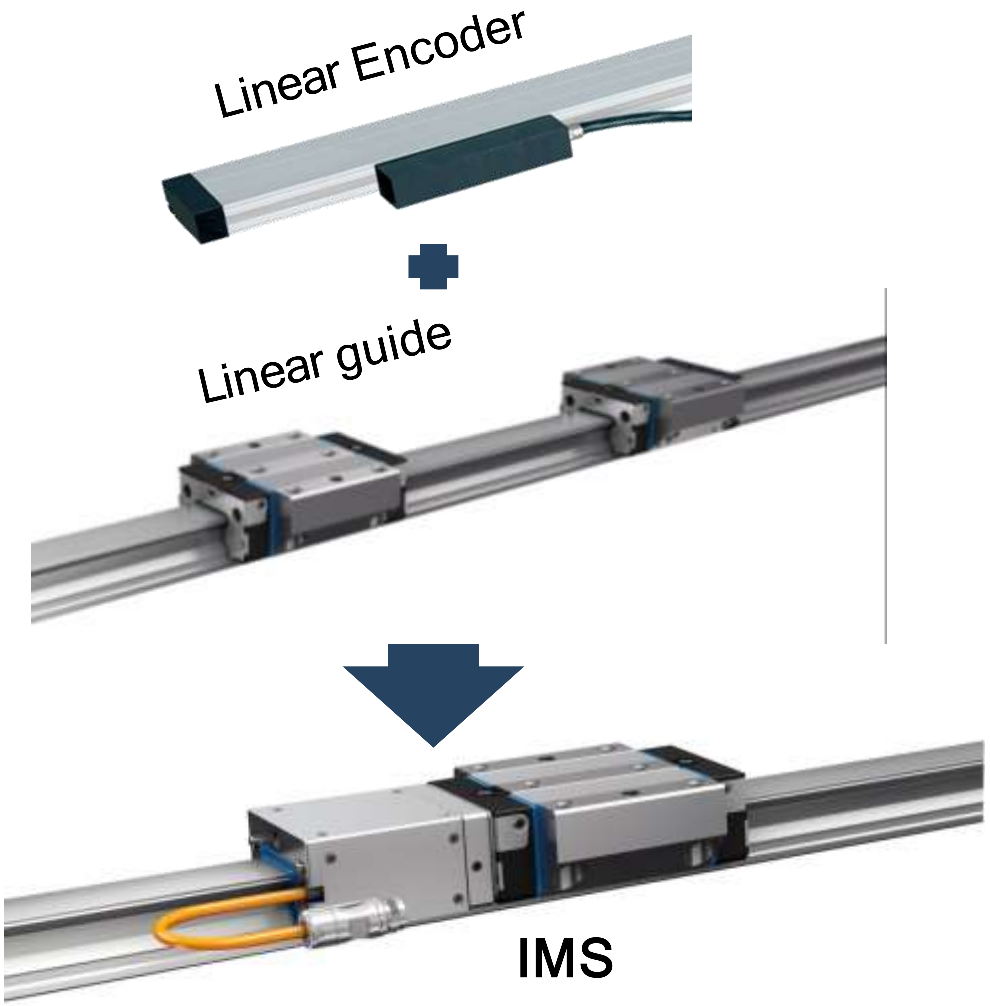
IMS-I: Integrated Measuring System - Incremental

Integrated Measuring System - IMS
What is IMS?

IMS is a ***high precise*** and accurate measuring system.

With its ***full integration*** into the guide rail customers have new opportunities for cost efficient and simpler machine designs.

Due to the ***inductive measuring principle*** IMS has a never before seen ***robustness and resistance*** in harsh environments.



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Integrated Measuring System - IMS
Components

Measuring Head (Scanner)

- Including sensors, electronics, connection cables and connector
- Ready-mounted on the runner block

Ball- or Roller-Runner-Block
(with mounted adapter plate)

Available in:
- different sizes
- different accuracy classes
- different preloading classes

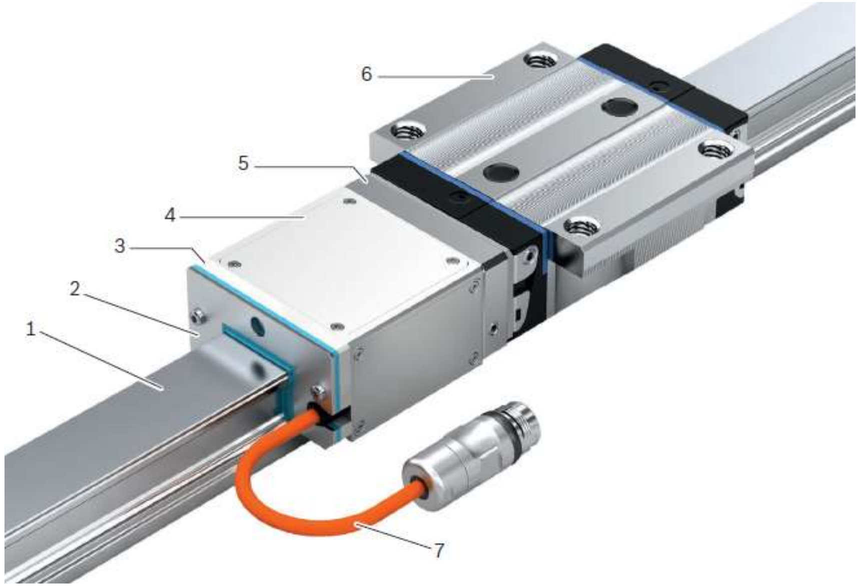
Guide Rail

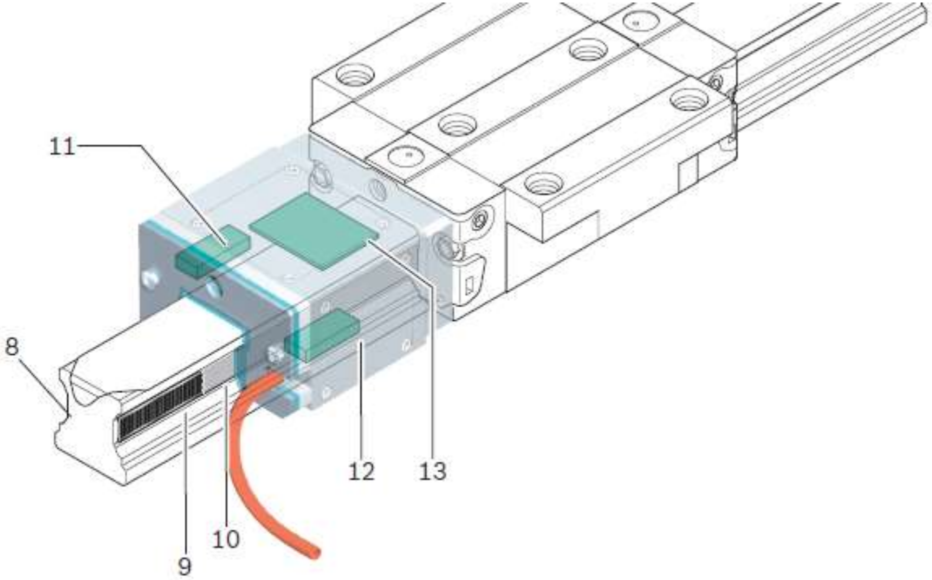
- with integrated, incremental scale
- with integrated absolute code band (IMS-A)
- with integrated reference marks (IMS-I)
- with cover strip or plastic/steel caps

The integration of the measuring system in the linear guide results in a **mechatronic system** which combines the functions of **guiding** mechanical loads **and measuring** of absolute positions in a single product.

External linear encoders (e.g. glass scales) are not needed anymore!

Integrated Measuring System - IMS
Design





1 Guide rail with scale, reference marks or absolute code band
2 End seal
3 Support plate
4 Scanner
5 Adapter plate (fixed to the runner block)
6 Runner block
7 Cable and connector

8 Reference marks (IMS-I) or absolute code band (IMS-A)
9 Incremental scale pitch
10 Scale protection via laser-welded stainless steel strip
11 Sensor for reference marks (IMS-I) or absolute code band (IMS-A)
12 Measuring sensor
13 Evaluation electronics

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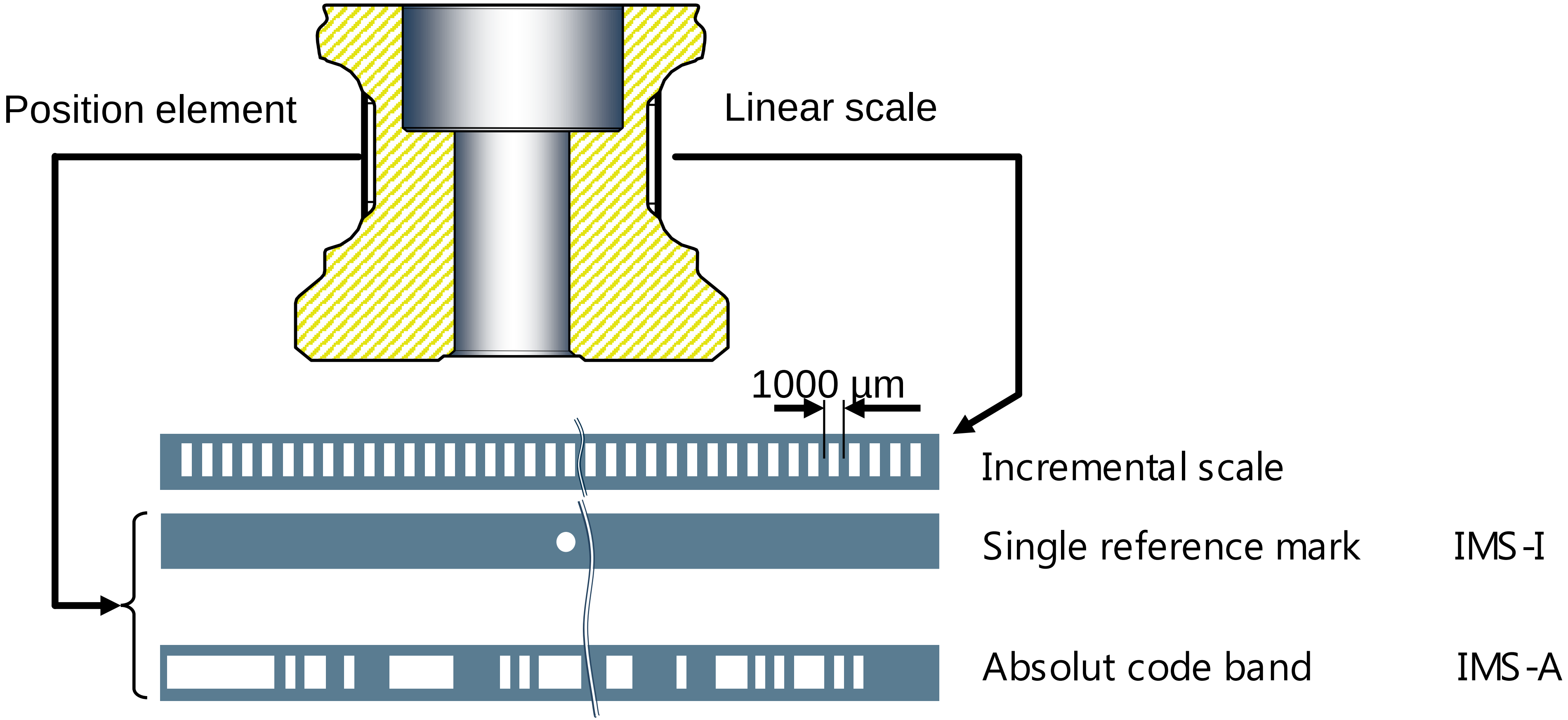
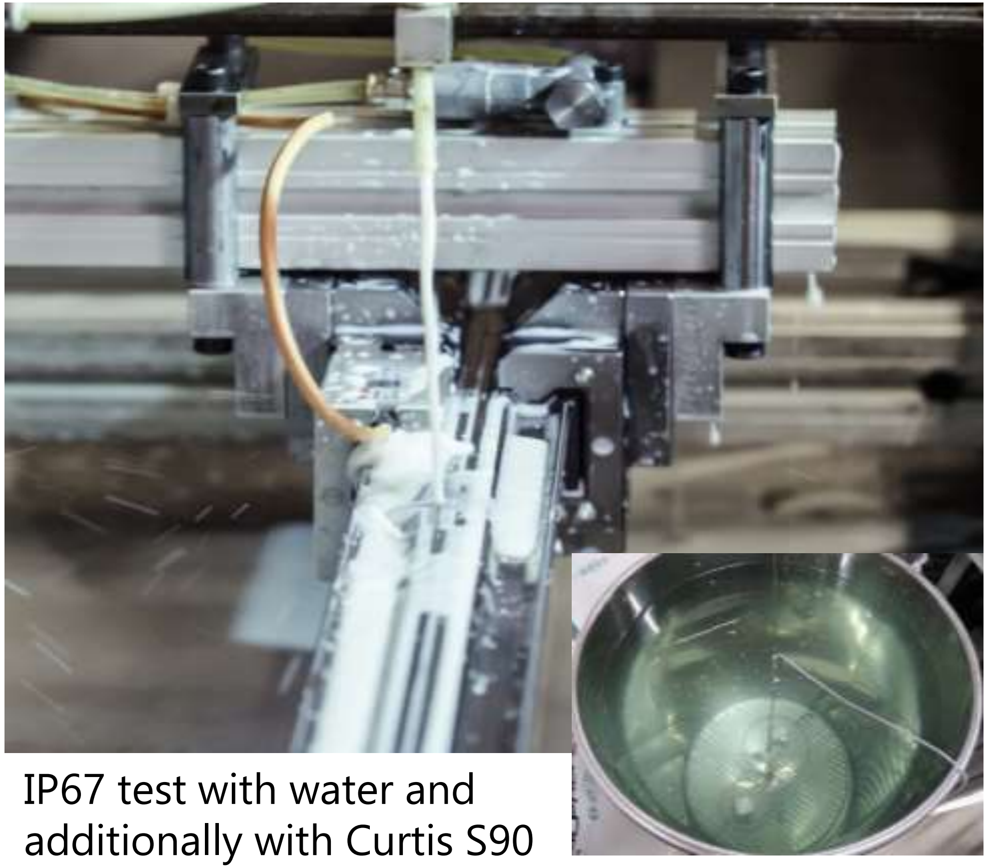
Design – IMS rail construction



Inductive measuring principle enables contactless position measurement.

The IMS measuring function is free of wear and maintenance, also under following conditions……..

Integrated Measuring System - IMS
Robustness



IP67 test with water and additionally with Curtis S90

IP67 dust-testing and additionally 800km in metal chip test drum

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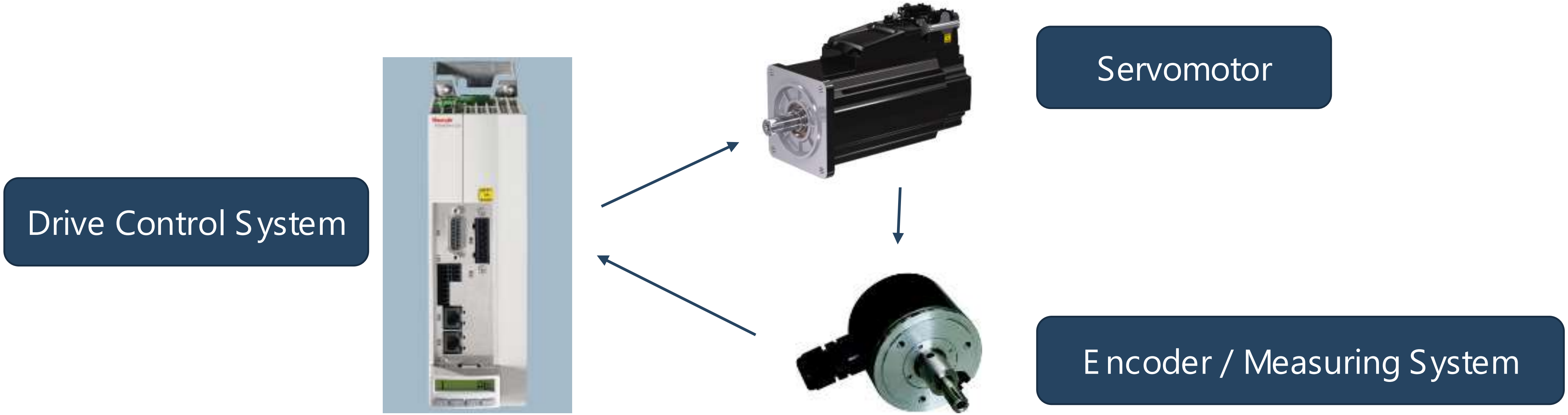
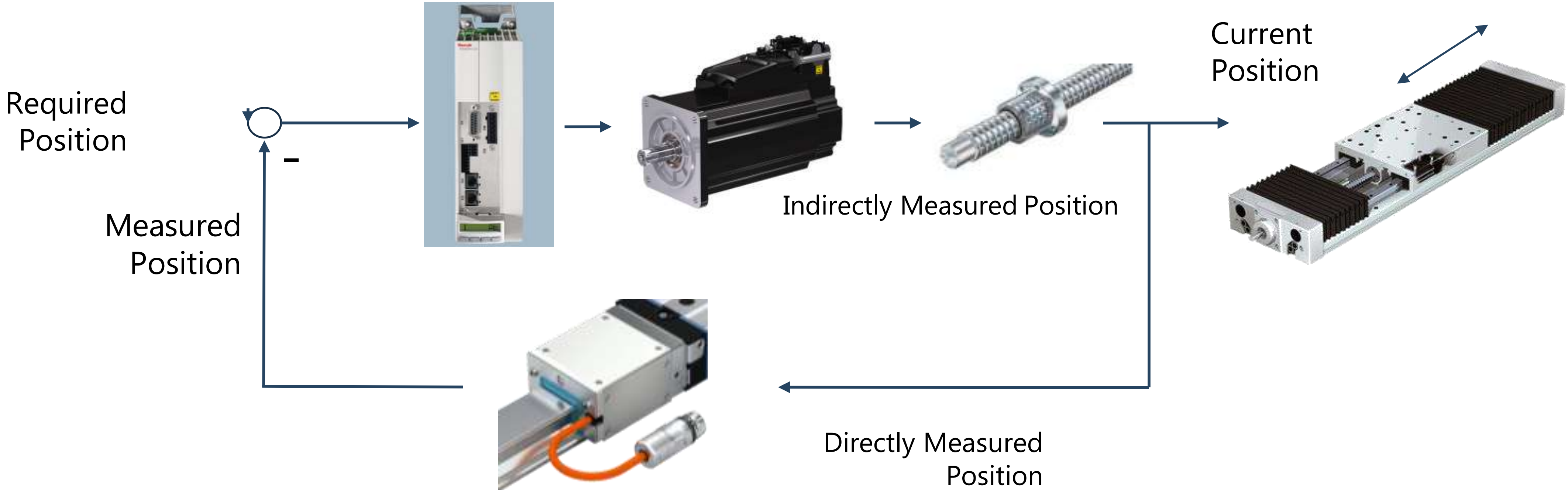
Why IMS?

For every **positioning task** the axis drive **servomotor** needs an additional **encoder**, which provides the actual position for the **drive control system**.



Integrated Measuring System - IMS
Why IMS?

Semi-Closed Loop Control <-> Closed Loop Control



Heidenhain-TV on YouTube: https://youtu.be/IZxeJ6sPaOw

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Why IMS?

Highlights:	Advantages:	Your Benefits:
Absolute measuring system	• absolute position information immediately after system power on • no battery required compared to competition	Power up without referencing movement minimizes machine run-up time and prevents damages to the tool and work piece
Inductive measuring principle	• wear- and maintenance-free • not sensitive to dirt and vibrations • not sensitive to EMC interference • no magnetic components	High level of reliability and reduced downtimes of your machines
Highly precise position ***resolution*** and ***repeatability***	• exact positioning • excellent control loop dynamics • very low path deviation	Excellent machine accuracy Excellent work piece quality
Complete integration into linear guide	• minimum assembly time • no adjustments for measuring system • very good protection against damage • high robustness and stiffness	Cost advantage due to time savings in terms of design, assembly, commissioning and service

By mounting the linear guide, the linear encoder for position measurement is already installed.

It is plug and play – no further measures are necessary!

Integrated Measuring System - IMS
Product Portfolio: Accuracies

Scale Accuracy	Up to ±3µm/m
Repeatability	± 0,25µm
Interpolation Accuracy	± 0,75µm
Position Value Resolution	Up to 0,025µm

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Product Portfolio: Interfaces IMS-A

HIPERFACE® by SICK

SSI SYNCHRON SERIELLES INTERFACE

M17 / 17-Pin

- IP67 protection

FANUC

M23 / 17-Pin

- IP67 protection
- Fits to standard FANUC connectors

DRIVE-CLiQ

- IP67 protection
- Fits SIEMENS MOTION-CONNECT accessories

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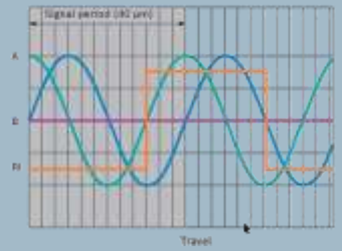
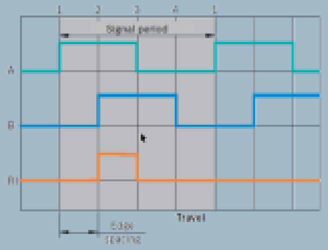
Product Portfolio: Interfaces IMS-I



1Vpp / 40µm



TTL 1µm,
TTL 5µm
TTL 10µm

M17 / 17-Pin

- IP67 protection

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Product Portfolio – Ball Rail System

FNS R1651

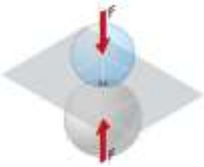
FLS R1653

SNH R1621

SNS R1622

SLS R1623

SLH R1624



Size:	**20 / 25 / 30 / 35 / 45**
Preload:	C1 / C2 / C3
Accuracy :	P / SP
Ball Chain:	without / with
Mounting Side:	right / left

Type:	**SNS – Standard**
Mounting:	from above
Accuracy:	± 3 µm/m and ± 5 µm/m @ 20°C
Length:	up to 4500 mm

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Product Portfolio – Roller Rail System

FNS R1851

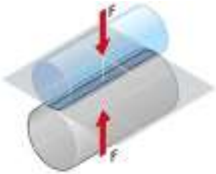
FLS R1853

SNH R1821

SNS R1822 [1]

SLS R1823 [1]

SLH R1824



Size:	35 / 45 / 55 / 65
Preload:	C2 / C3
Accuracy:	P / SP
Mounting Side:	right / left

Type:	SNS – Standard
Mounting:	from above
Accuracy:	± 3 µm/m and ± 5 µm/m @ 20°C
Length:	up to 4000 mm

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Integrated Measuring System - IMS
Product Portfolio – Connection- and Extension Cables

Standard Length:
0,5m 2m, 3m, 5m,
8m, 10m, 15m, 20m

Connection-Cable Rexroth
RKG0055
IndraDrive with EC Interface (12VDC)

Connection-Cable SIEMENS
RKG0071
Sensor Modules SME 25 / SME 125

Extension-Cable:
RKG0057

RKG0058, open ends

Single Connector
RGS1711

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Integrated Measuring System - IMS
Product Portfolio – Technical Data

Parameter	Value		Notes
Voltage supply:	V_{DD}: 4,75 …. 24 V DC I_{MAX}: 300 mA		Overvoltage protection: ≤ 30V DC
Operating temperature range:	0 °C …. 50 °C		
Temperature range for storage / transport:	-10 °C … 70 °C		
Measuring speed:	5 m/s		Additional dependency of guide element!
Protection type:	IP 67		(tested with Curtis S90 cooling lubricant)
Maximum acceleration:	500 m/s^2		
Shock:	500 m/s^2 / 11 ms		According to EN 60068-2-27:1993 / IEC 68-2-6:1995
Vibration:	100 m/s^2		55-2000Hz, according to EN 60068-2-6:1996 / IEC 68-2-6:1995
Electromagnetic compatibility:	Immunity to interference: Emitted interference:	EN 61326-1: 2006 EN 61000-6-2, Class B	CE mark RoHS and UL conform

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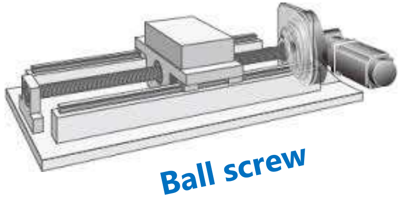
IMS is the most accurate integrated measuring system available on market.

The system accuracy meets the specification of linear encoders with glass scales.

Integrated Measuring System - IMS
Where to use?

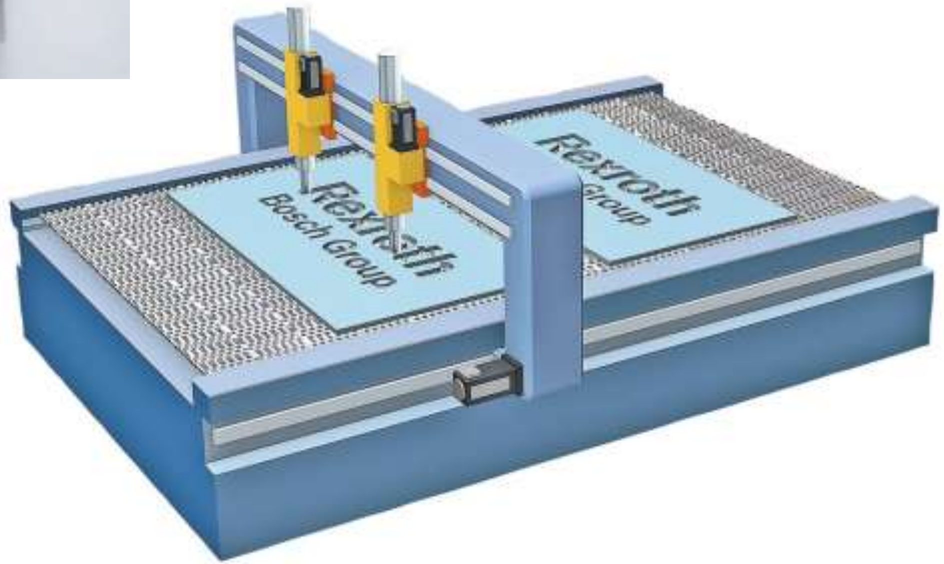
Kind of drive chains IMS is used for:



Integrated Measuring System - IMS
Where to use?

2D & 3D Laser cutting machines:

- Linear motor in X- and Y-axis
- X-axis as gantry
- Y-axis carries laser cutters



➡ **IMS-I and IMS-A systems are used in many different machine types from different laser cutting machine builders.**

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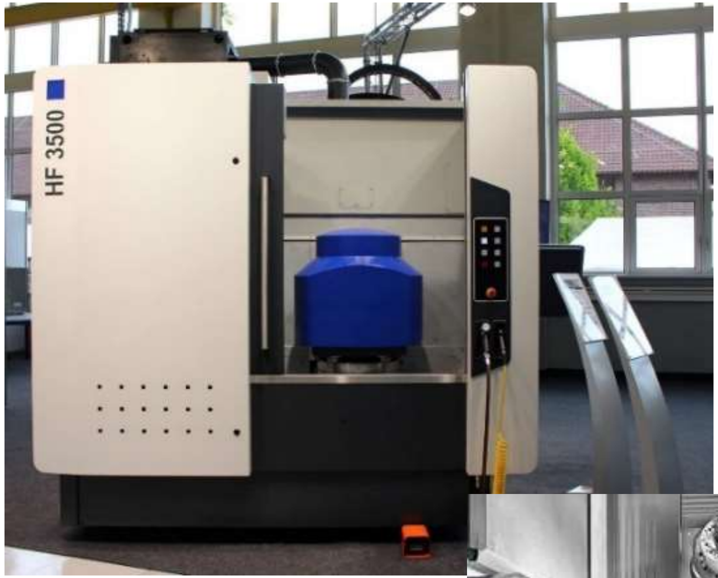
Integrated Measuring System - IMS
Where to use?



Machine Tool

Heller – DE
www.Heller.biz

5-Axis Machining Centre HF

<u>New machine series for 2017:</u>
HF 3500 / 5500

RSF Size 45 and 35
IMS is used for X-, Y- and Z-axis -> 3 IMS Systems per machine

Drive chain: ball screw assembly

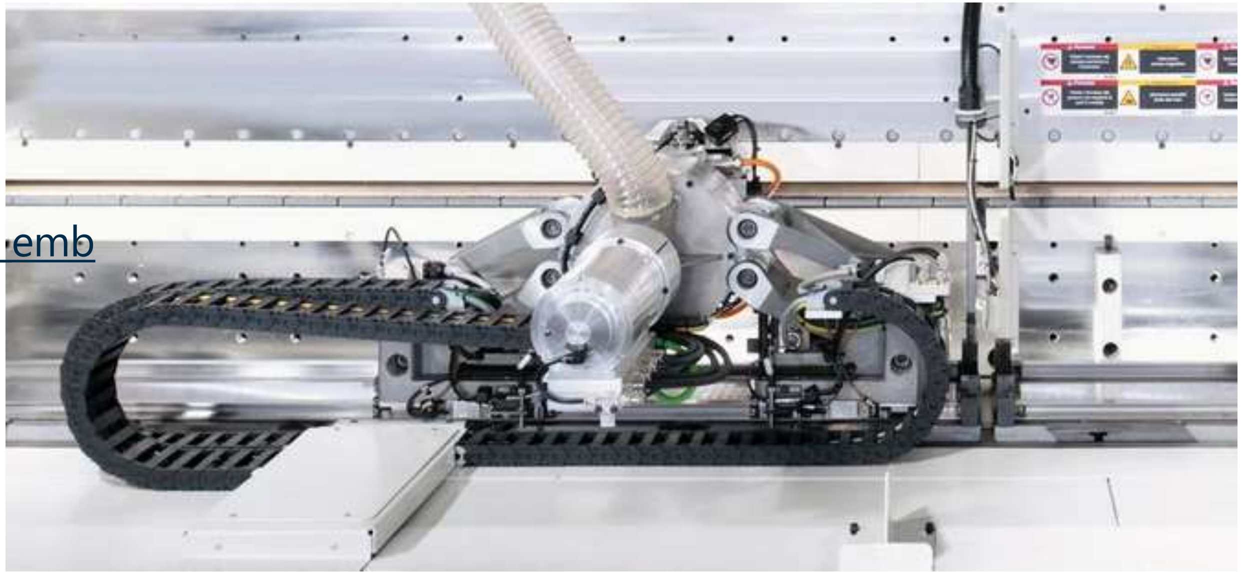
Integrated Measuring System - IMS
Where to use?

Wood-Working Industry

Edge banding machine COMBINA

Edge processing unit KFA x50
(video:
https://www.youtube.com/watch?feature=player_embedded&v=Tqb5lo5M9p0#t=139
-> IMS from 2:17 to 2:33)

Drive chain: linear motor



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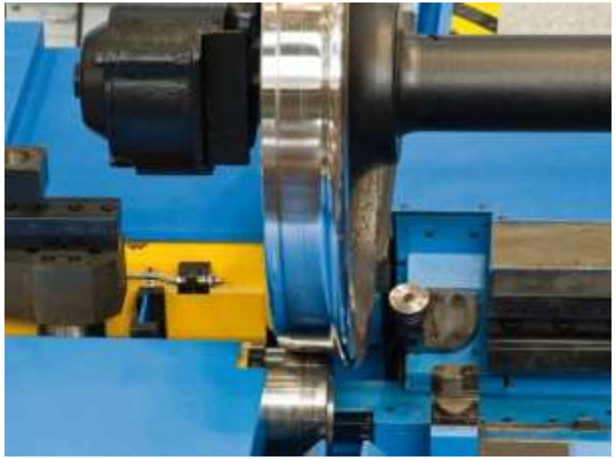
Where to use?

Wheelset machining – Underfloor wheelset lathe

IMS systems for measuring wear and the diameter of wheel.



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Integrated Measuring System - IMS
Where to use?

Automation systems for battery and display manufacturing systems

Example: modular system, based on Rexroth comp

- IMS-I and IMS-A: BSHP size 25
- Ball rail system BSHP
- IndraDrive Cs
- Linear motor MCP/MCS040

Drive chain: linear motor

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Integrated Measuring System - IMS
Where to use?

Plotter Application:

- Linear motor in X- and Y-axis
- X-axis is a gantry configuration
- Y-axis carries the plotter

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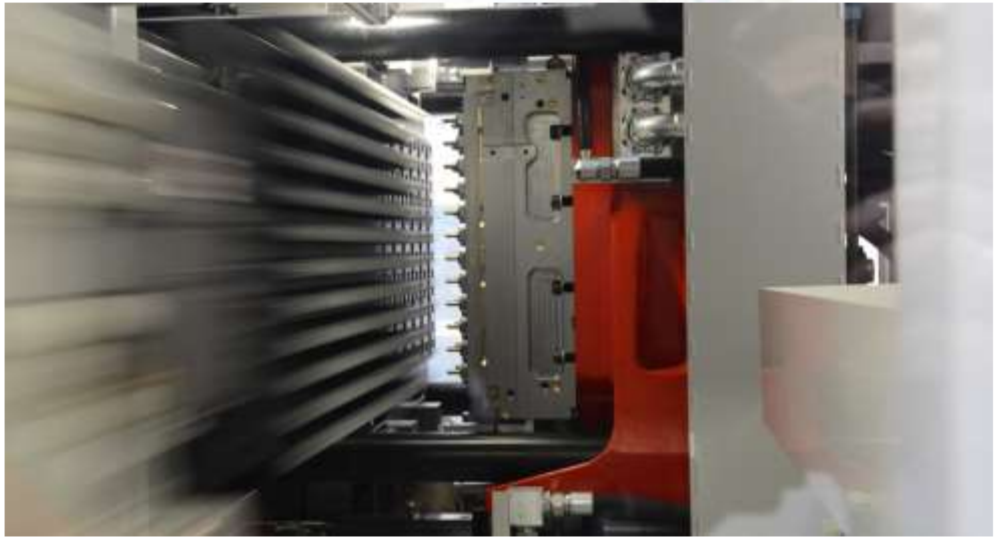
Where to use?

PET Preform Injection systems

Machine-Type: IPS 220, IPS 400

The take-out robot is driven by latest-generation belt-free (and, therefore, maintenance-free) linear motors, ensuring accuracy, speed and repeatability of robot movements.



- *IMS-I ball rail system, Size 45*
- *Rexroth linear motor MLP 100C-0190 and IndraDrive*

Video:
http://multimedia.sacmi.com/ABCENetMediaPlayer.aspx?src=/Streaming/SacmiImola/Plastics/Injection_Preforms_System.wmv&AutoPlay=1
-> *take out robot with linear motor @ ~ 1min, 36s*

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Where to use?

Welding technology and automation
-> IMS used for quality control of process.

Process:
CD-welding with high-current pulse

Typical applications:
Gear box parts, common rail, airbag inflator, seat recliner, etc.

IMS-A, roller rail system, size 35

Drive chain: ball screw assembly

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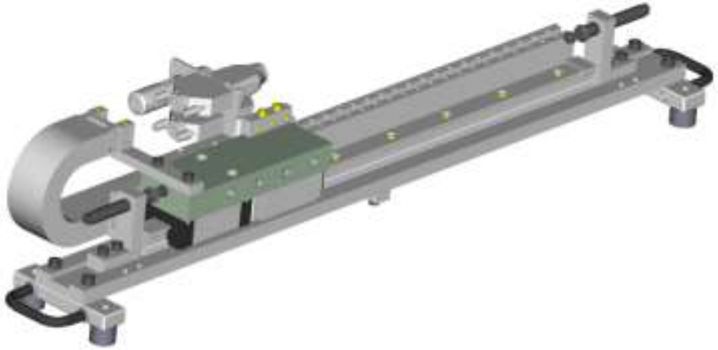
Where to use?

Demonstration unit from Bosch Rexroth:

Small axis consists of:

- Ironless linear motor
- IMS-I / IMS-A runner block and rail, BSHP size 20
- Mechanical assembly



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IMS can be used for many different applications in different branches.

Integrated + Precise + Robust

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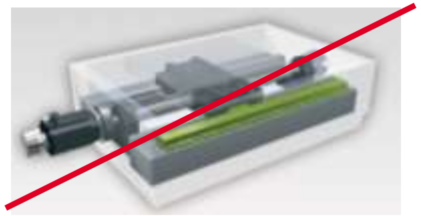
Integrated Measuring System - IMS
TCO - total cost of ownership

Space Saving

-> Allows simple machine designs due to the full integration

-> No additional space required

-> Elimination of stop and mounting surfaces

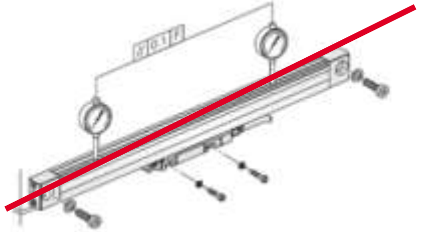


Easy Installation

-> Minimum mounting time: the guide assembly is mounted at the same time as the measuring system

-> No adjustment of the measuring system

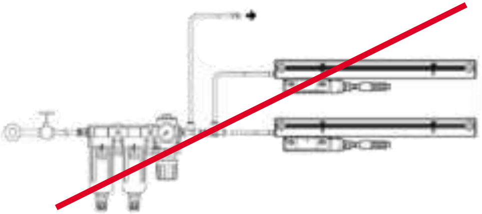
-> No need for additional components



Protection Class

-> Elimination of all sealing air components and installations like hoses, pressurized air and their service (filter change)

-> Very good protection against damage



TCO Calculation Tool

-> Usage of IMS leads to savings in designing, installation and service purposes

-> Savings can be calculated with the TCO Tool

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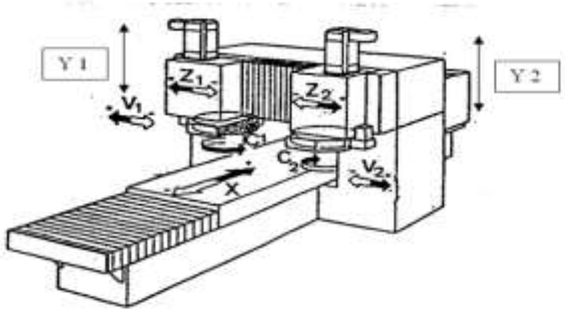
Integrated Measuring System - IMS
TCO - total cost of ownership



Example:

Grinding machines for profiled guide rails in Schweinfurt plant

Problem:

Sealed linear encoders (glass scale) cause downtime of machine and regular cost for maintenance (cleaning glass scale) due to contamination from grinding process.

Solution:

Retrofitting old grinding machines by using IMS systems instead of sealed linear encoders.

Benefits:

Cost savings of approx. 10K€ per machine due to reduced downtime of machine and no maintenance for IMS.

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Thank you!